

SPECIFICATION OF ACOUSTIC PERFORMANCE OF ACOUSTICAL DOORS

I GENERAL

A. Description

1. Related Documents: All provisions of this contract and Drawings.
2. Work Included: Provide all-steel sound insulating doors, frames, seals, automatic door bottoms, gaskets, appropriate closure/locking hardware and glazing. All components to be products supplied by the single manufacturer, and shall comply with specified laboratory acoustical ratings as tabulated at end of this section.
3. Door manufacturer's representative to be present during installation and final check-out by the Project Manager.

B. Quality Assurance

1. Use adequate numbers of skilled workmen who are thoroughly trained and experienced in the necessary crafts and who are completely familiar with the specified requirements and the methods needed for proper performance of the work of this section.
2. Use a subcontractor, under the supervision of the door manufacturer, who has been engaged in the manufacture of acoustical door and window systems for at least five years immediately prior to the start of this Work, and who has a history of successful production acceptable to the Project Manager.
3. Except as otherwise accepted in advance by the Project Manager, provide work of this Section designed and furnished by one Contractor.
4. In addition to complying with pertinent requirements of governmental agencies having jurisdiction, comply with pertinent recommendations contained in the National Association of Architectural Metal Manufacturers:
 - a. "Hollow Metal Technician and Design Manual" (1977); and
 - b. "Fire Rated Hollow Metal Doors and Frames", 2nd edition, 1983.
 - c. "Glazing Manual" of the Flat Glass Marketing Association.
5. Contractor will be responsible for all details associated with the door and closure/locking hardware to achieve acoustical performance; including full perimeter of door and frame.

C. Submittals

1. Product Data: Submit certification of Sound Transmission Class (STC) rating, along with transmission loss values within individual 1/3 octave band frequencies, for all acoustical doors and windows performed by an accredited independent acoustical laboratory.
 - a. All testing shall be in accordance with ASTM E90 latest version.
2. Product cutsheet and job reference.
3. Field performance data tested by an independent laboratory or consultant, in accordance with ASTM E336, shall be submitted showing compliance with specified values.
4. Shop Drawings: Submit shop drawings for fabrication and installation of all acoustical doors, frames, seals, gaskets, closure hardware and windows.
5. Identification: Provide a schedule identifying location and corresponding details of products.
6. Letter of field observation that partition framing is suitable for the installation and proper operation and performance of the door systems.
7. Fabrication: No acoustical door, frame or window shall be fabricated until submittal has been issued and accepted.

D. Product Handling

1. Deliver acoustical doors and windows crated to provide protection during storage on site.
2. Inspection: All components shall be thoroughly inspected for damage prior to installation. Minor repairs may be repaired on site upon acceptance, otherwise complete replacement is necessary.

E. Guarantee

1. Upon completion of work of this Section, and as a condition of its acceptance, deliver to the Project Manager two copies of the manufacturer's standard written two-year guarantee.

II PRODUCTS

A. Acceptable Manufacturers

1. Manufacturer: Subject to compliance with requirements, provide specified products of one of the following:

- a. Yokoh International Limited
- b. Industrial Acoustics Company
- c. NAP Acoustics
- d. Approved equivalent

B. Door Assembly

Acoustical Solid Door

1. Frames: Frames shall be fabricated from 14 ga. (2.0mm) steel with mitered, welded and ground smooth corners. Strike, hinge and hardware reinforcement shall be not less than 3/16" (4.8mm) thick steel.
2. Doors: Doors shall be formed 16ga./14ga. (1.6mm/2.0mm) steel face sheets joined together by appropriate inner reinforcing. The core shall be fully packed with an incombustible, non-coupling filler.
3. Windows: Glazing and seals shall be as required to comply with acoustical performance specifications.
4. Acoustical Seals: Full perimeter acoustical seals shall be provided from the single manufacturer as necessary to achieve specified laboratory and field acoustical ratings. Head and jamb seals shall be magnetic.

C. Sill Conditions

1. Door manufacturer shall provide necessary sill material or condition in order to achieve acoustical ratings; or advise the Project Manager or Contractor of required detail if different than shown on contract documents.

D. Closure and Locking Hardware

1. Door manufacturer shall provide all operating, closure and locking hardware necessary to achieve highest field installed acoustical rating, and meet applicable codes and regulations.
2. Door operating hardware shall be accepted by the Project Manager.

E. All materials are to comply with pertinent regulations and requirements of all governing agencies having jurisdiction.

III EXECUTION

A. Installation

1. The entire installation shall be done, or at least supervised, by the manufacturer or their representative.
2. The door manufacturer shall inspect and accept all door openings prior to installation of door assemblies.

3. Install acoustical doors and windows in accordance with manufacturer's printed instructions and recommendations by a qualified installer.
4. Cut and trim framing during installation only with the acceptance of manufacture and in accordance with manufacturer's recommendations.
5. Restore finish and remove and replace members as directed where cutting and trimming has impaired strength or appearance.
6. Do not install members which are warped, bowed, deformed, or otherwise damaged or defaced to such extent as to impair strength, appearance or performance. Remove and replace members, as directed, which have been damaged during installation.
7. Grout fill jambs and head with masonry grout material when installed in masonry partitions, and fully pack with batt insulation when installed within drywall/stud partitions.

IV FINAL INSPECTION/ACOUSTICAL PERFORMANCE

A. Inspection

1. Pay such costs as are required to secure job site representation by the manufacturer of the acoustical door and window systems. At completion of the project the manufacturer shall submit letter to Employer that installation and adjustment is complete and in accordance with accepted Shop Drawings and Contract Documents.
2. Field Certification: The Contractor shall guarantee acoustical performance in the field of the products. The performance rating shall be as follows:

Door Acoustical Performance		
Door	Laboratory Ratings (ASTM E90)	Field Ratings (ASTM E336)
	Minimum STC ⁽¹⁾	NIC ⁽²⁾
<u>Acoustical Solid Door</u> - Conference Rooms - Refer to door schedules.	50	43

⁽¹⁾Sound Transmission Class

⁽²⁾Noise Isolation Class

Tested nominally 1.5m from each side of door.

3. After installation of the acoustic door is completed and to the satisfaction of the door manufacturer or his designated representative, initial field test shall be conducted by an independent testing agency employ by the Contractor. The Contractor shall be on hand to observe the field testing and to make door adjustments as required. Cost of the field tests will be the responsibility of the Contractor.

4. Field tests to be conducted in accordance with E336 for Noise Reduction and the NIC rating will be calculated in accordance with ASTM E413.
5. All corrections required to comply with the specified field ratings shall be the responsibility of the Contractor, along with subsequent tests.
6. Any re-testing required due to product installation not meeting the acoustical performance shall be performed and the cost will be the responsibility of the Contractor.

END OF SECTION